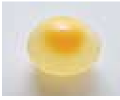
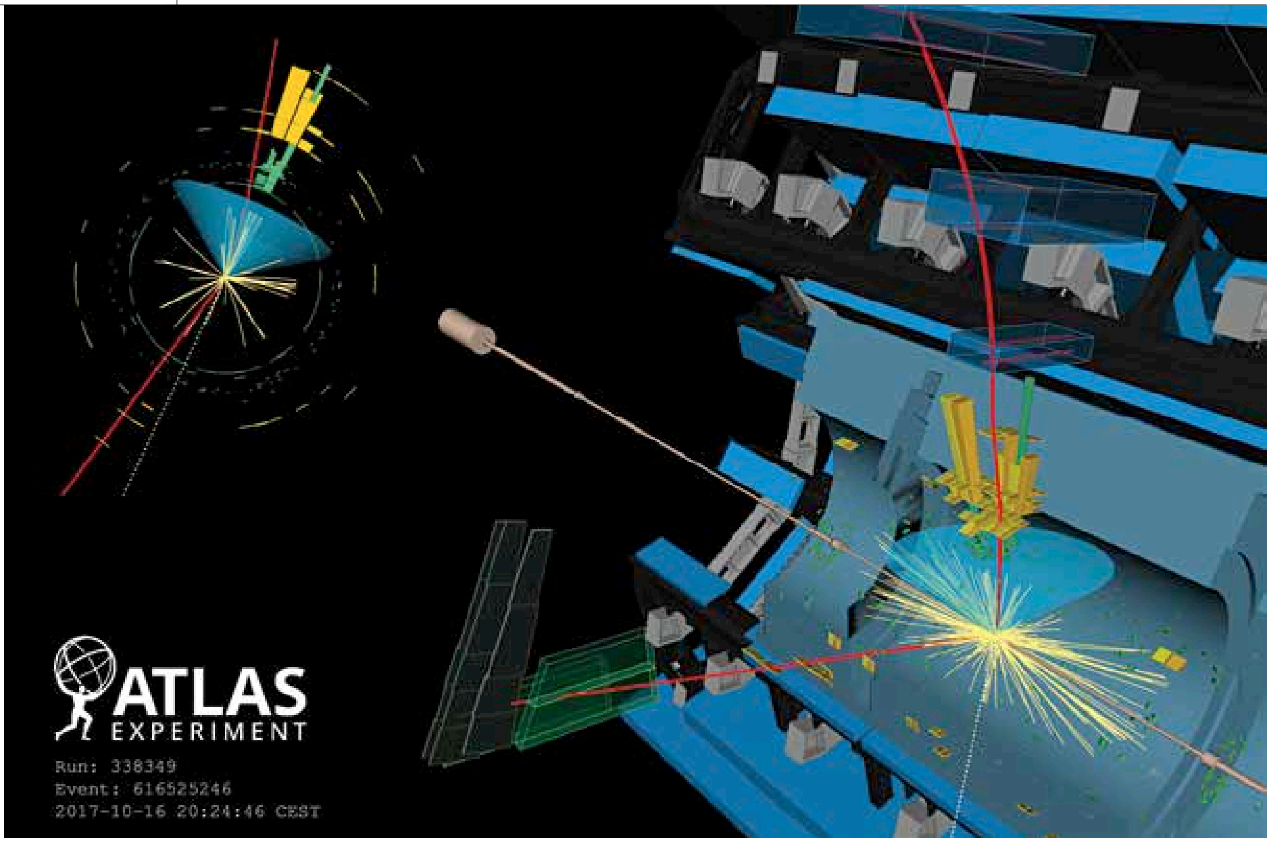


Warped egg yolks
Scientists experiment on egg yolk to measure soft matter deformation and understand concussions. **http://dgit.in/feb21-21**

"Wrinkled" graphine
Researchers develop new graphine nanochannel water filter by microscopically wrinkling its surface. **http://dgit.in/feb21-22**



Credit: Cern

The measurement era

What we have learned about the Higgs boson since its discovery

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Just after the Big Bang, all particles in the universe were just flying around at the speed of light. Suddenly, a universal field snapped over, around and within the universe, giving particles their mass. This is known as the Higgs field, which used the Higgs boson to provide mass to other particles. The particles with mass were suddenly moving at much slower speeds than the universe around them, and it is this early clumping of mass that eventually gave rise to all the large scale structures made up of galaxies and vacuum seen within the universe such as walls, filaments, rings, galactic superclusters and voids. There is dark matter in there too, but that has not been seen, and neither has the Higgs boson.

The Higgs boson has not been directly "observed" yet. The debris of events are captured in signatures, and scientists can spot only the decayed particles that are detected after energetic interactions that last for extremely short periods of time. The cluster of particles detected after a Higgs boson decays can also be caused by a number of other particle interactions. Finding out whether or not a detected event involves a Higgs boson or not is very similar to the forensic accounting techniques used to detect money laundering (that you can read about in this month's dmystify). Instead of looking at every transaction, they look at transactions at scale to identify the ones that are making suspiciously large amounts of profit. Scientists looked at the signatures of thousands of events, accounted for the ones that were predicted to occur, and then single out the events that are very likely to involve a Higgs boson. This is why with every announcement of a discovery, CERN scientists provide a statistical confidence rating for the prediction.

BEAUTY OF THE HIGGS BOSON

The announcement of the discovery of the Higgs boson in 2012, was for example given a 5 sigma rating. Over the years, the ATLAS instrument at CERN has detected 30 percent of the decays predicted by the standard